

Ro.74A | 混合 heat covariance 的局部 size lemma

core 由局部能量支付, exterior 由两个显式 Gaussian 二次尾支付

四块正 majorization 给出 $\theta^{1/4}$ 或 θ 付款; 旧的 Ro.73X exterior package 单独不够, 新 velocity endpoint tail 与复用的 gradient tail 只保证有限, 不保证小或可吸收。

01 / CANONICAL REPORT

Frozen definitions and the clock qualification

The standard and viscosity-adapted time intervals frozen in Ro.73X are

$$I_R^{\text{std}} = (t_0 - R^2, t_0), \quad I_R^\nu = (t_0 - R^2/\nu, t_0). \quad (1.1)$$

Both scale parabolically because ν is unchanged by the Navier--Stokes scaling. They are not interchangeable.

For either clock, the canonical local energy frozen in Ro.73X is

$$\mathcal{E}^\square(z_0, \rho) = \frac{1}{\rho} \text{ess sup}_{t \in I_\rho^\square} \int_{B_\rho} |u|^2 dx + \frac{\nu}{\rho} \int_{I_\rho^\square} \int_{B_\rho} |\nabla u|^2 dx dt, \quad \square \in \{\text{std}, \nu\}. \quad (1.2)$$

Every comparison below uses the same $\square$ on the left and right. No assertion replaces I_R^ν by a standard interval, or conversely.

Because the right-hand side uses radius $4R$, the common quantifier is

$$I_{4R}^\square \Subset (0, T). \quad (1.3)$$

Let $P_s = e^{s\Delta}$ be the periodic heat semigroup. At almost every physical time for which $u(t) \in H^1(\mathbb{T}^3)$, set

$$\gamma_m(\theta) = \theta^{-2} \exp\left(-\frac{4^{m-1}}{32\theta}\right). \tag{2.3}$$

For almost every $t \in I_R^\square$, define

$$\begin{aligned} U_\gamma(t; R, \theta) &= \sum_{m \geq 1} \gamma_m(\theta) \int_{A_m(R)} |\tilde{u}(t, y)|^2 dy, \\ G_\gamma(t; R, \theta) &= \sum_{m \geq 1} \gamma_m(\theta) \int_{A_m(R)} |\nabla \tilde{u}(t, y)|_F^2 dy. \end{aligned} \tag{2.4}$$

The scale-invariant exterior quantities used below are

$$\begin{aligned} \mathcal{U}_{\text{ext}}^{\infty, \square}(z_0, R; \theta) &= \operatorname{ess\,sup}_{t \in I_R^\square} \frac{U_\gamma(t; R, \theta)}{R}, \\ \mathcal{G}_{\nabla, \text{ext}}^{1, \square}(z_0, R; \theta) &= \frac{\nu}{R} \int_{I_R^\square} G_\gamma(t; R, \theta) dt. \end{aligned} \tag{2.5}$$

The superscripts record the time exponents, not derivative orders. The second row is not a new tail: comparison with Ro.73X (7.2) gives the exact identity

$$\mathcal{G}_{\nabla, \text{ext}}^{1, \square}(z_0, R; \theta) = \mathcal{D}_{\text{ext}}^\square(z_0, R; \theta). \tag{2.5a}$$

Only $\mathcal{U}_{\text{ext}}^{\infty, \square}$ is newly introduced in this note. We use $\mathcal{D}_{\text{ext}}^\square$ from now on.

Lemma 2.1 — scaling and finiteness of the tails

Under

$$u_\lambda(t, x) = \lambda u(\lambda^2 t, \lambda x), \qquad R_\lambda = R/\lambda, \tag{2.6}$$

with the corresponding rescaled center and interval,

$$\mathcal{U}_{\text{ext}}^{\infty, \square}[u_\lambda] = \mathcal{U}_{\text{ext}}^{\infty, \square}[u], \qquad \mathcal{D}_{\text{ext}}^\square[u_\lambda] = \mathcal{D}_{\text{ext}}^\square[u]. \tag{2.7}$$

Both quantities are finite under (1.7).

Proof

On corresponding annuli,

$$\int |u_\lambda|^2 dx = \lambda^{-1} \int |u|^2 dx, \quad \int |\nabla u_\lambda|^2 dx = \lambda \int |\nabla u|^2 dx. \quad (2.8)$$

Together with $R_\lambda^{-1} = \lambda R^{-1}$ and $dt_\lambda = \lambda^{-2} dt$, this proves (2.7), with the periodic lattice rescaled in the standard local Navier--Stokes scaling convention.

Each lifted annulus contains at most $C[1 + (2^m R)^3]$ fundamental cells. Hence

$$\begin{aligned} U_\gamma(t) &\leq C \|u(t)\|_{L^2(\mathbb{T}^3)}^2 \sum_{m \geq 1} \gamma_m [1 + (2^m R)^3], \\ G_\gamma(t) &\leq C \|\nabla u(t)\|_{L^2(\mathbb{T}^3)}^2 \sum_{m \geq 1} \gamma_m [1 + (2^m R)^3]. \end{aligned} \quad (2.9)$$

The sums converge because γ_m decays faster than geometrically. Now (1.7) proves finiteness. $\square$

03 / CANONICAL REPORT

Positive core/exterior majorization

For $x \in B_R$ and $0 < s \leq \theta R^2$, define

$$\begin{aligned} U_c &= \int_{C_R} g_s(\tilde{x} - y) |\tilde{u}(y)|^2 dy, & U_e &= \int_{E_R} g_s(\tilde{x} - y) |\tilde{u}(y)|^2 dy, \\ G_c &= \int_{C_R} g_s(\tilde{x} - y) |\nabla \tilde{u}(y)|_F^2 dy, & G_e &= \int_{E_R} g_s(\tilde{x} - y) |\nabla \tilde{u}(y)|_F^2 dy. \end{aligned} \quad (3.1)$$

These are positive pieces of the uncentered moments. They are not separate covariances. From (1.5),

$$\boxed{D_s \sqrt{k_s} \leq \frac{1}{\sqrt{2}} \left(G_c \sqrt{U_c} + G_c \sqrt{U_e} + G_e \sqrt{U_c} + G_e \sqrt{U_e} \right).} \quad (3.2)$$

This is the core/core, core/exterior, exterior/core, and exterior/exterior split used in the theorem.

There is no claimed exact four-term covariance identity. Directly splitting $u = 1_{C_R} u + 1_{E_R} u$ inside a covariance creates cross terms such as

$$-2(P_s(1_{C_R} \nabla u)) : (P_s(1_{E_R} \nabla u)), \quad (3.3)$$

and the square root of the velocity covariance cannot be expanded linearly. The positive majorization (3.2) avoids that false identity.

The Ro.73X Gaussian annulus lemma gives, uniformly in the full scale interval,

$$\boxed{U_e(t,x,s)\leq CR^{-3}U_\gamma(t),\qquad G_e(t,x,s)\leq CR^{-3}G_\gamma(t).}\tag{3.4}$$

Also,

$$\int_{B_R}G_c(t,x,s)\,dx\leq\int_{B_{2R}}|\nabla u(t,y)|_F^2\,dy.\tag{3.5}$$

Finally, periodic heat ultracontractivity implies

$$\|U_c(t,\cdot,s)\|_{L^\infty(B_R)}^{1/2}\leq C_{\mathbb{T}^3}s^{-3/4}\|u(t)\|_{L^2(B_{2R})}.\tag{3.6}$$

04 / CANONICAL REPORT

The localized size theorem

For $I=I_R^\square$, put

$$\begin{aligned} A_c(I,R)&=\frac{1}{R}\operatorname{ess\,sup}_{t\in I}\int_{B_{2R}}|u(t)|^2\,dx,\\ B_c(I,R)&=\frac{\nu}{R}\int_I\int_{B_{2R}}|\nabla u|_F^2\,dx\,dt. \end{aligned}\tag{4.1}$$

Theorem 4.1 — core/exterior Gaussian size lemma

Assume (1.6)–(1.7). Then

$$\boxed{\mathcal{K}_D^\square(z_0,R;\theta)\leq C_{\mathbb{T}^3}\big[\theta^{1/4}A_c^{1/2}B_c+\theta B_c(\mathcal{U}_{\operatorname{ext}}^{\infty,\square})^{1/2}+ \theta^{1/4}A_c^{1/2}\mathcal{D}_{\operatorname{ext}}^\square+\theta(\mathcal{U}_{\operatorname{ext}}^{\infty,\square})^{1/2}\mathcal{D}_{\operatorname{ext}}^\square\big]}.\tag{4.2}$$

Since $0<\theta\leq 1$, this implies the factorized bound

$$\mathcal{K}_D^\square \leq C_{\mathbb{T}^3} \theta^{1/4} \left(A_c^{1/2} + (\mathcal{U}_{\text{ext}}^{\infty, \square})^{1/2} \right) (B_c + \mathcal{D}_{\text{ext}}^\square). \quad (4.3)$$

For either clock, the matching Ro.73X local energy gives

$$\boxed{\mathcal{K}_D^\square(z_0, R; \theta) \leq C_{\mathbb{T}^3} \theta^{1/4} \left[\mathcal{E}^\square(z_0, 4R) + \mathcal{U}_{\text{ext}}^{\infty, \square}(z_0, R; \theta) + \mathcal{D}_{\text{ext}}^\square(z_0, R; \theta) \right]^{3/2}.} \quad (4.4)$$

Here $\mathcal{K}_{D,cc}$ denotes the contribution obtained from $G_c \sqrt{U_c} / \sqrt{2}$ in (3.2).

Proof: core--core

By (3.5)--(3.6),

$$\int_{B_R} G_c \sqrt{U_c} \, dx \leq C s^{-3/4} \|u(t)\|_{L^2(B_{2R})} \|\nabla u(t)\|_{L^2(B_{2R})}^2. \quad (4.5)$$

Since

$$\int_0^{\theta R^2} s^{-3/4} \, ds = 4\theta^{1/4} R^{1/2}, \quad (4.6)$$

we obtain

$$\mathcal{K}_{D,cc}^\square \leq C \theta^{1/4} A_c^{1/2} B_c. \quad (4.7)$$

For either fixed $\square$, the spatial inclusion $B_{2R} \subset B_{4R}$ and the matching time inclusion $I_R^\square \subset I_{4R}^\square$ give

$$A_c + B_c \leq 4\mathcal{E}^\square(z_0, 4R). \quad (4.8)$$

Thus the core--core row is paid by $C\theta^{1/4}(\mathcal{E}^\square)^{3/2}$. Combining (4.8) with (4.3) and the elementary bound $(a^{1/2} + b^{1/2})(c + d) \leq C(a + b + c + d)^{3/2}$ proves (4.4).

Proof: core--exterior

Using (3.4)--(3.5),

$$\int_{B_R} G_c \sqrt{U_e} \, dx \leq CR^{-3/2} U_\gamma(t)^{1/2} \int_{B_{2R}} |\nabla u(t)|^2 \, dx. \quad (4.9)$$

The integrand no longer depends on s . Integrating over a scale interval of length θR^2 , and then over physical time, gives

$$\mathcal{K}_{D,ce}^\square \leq C\theta B_c(\mathcal{U}_{\text{ext}}^{\infty,\square})^{1/2}. \quad (4.10)$$

Proof: exterior--core

From (3.4) and $|B_R| \simeq R^3$,

$$\int_{B_R} G_e \sqrt{U_c} \, dx \leq Cs^{-3/4} G_\gamma(t) \|u(t)\|_{L^2(B_{2R})}. \quad (4.11)$$

Using (4.6) and the definitions (4.1), (2.5),

$$\mathcal{K}_{D,ec}^\square \leq C\theta^{1/4} A_c^{1/2} \mathcal{D}_{\text{ext}}^\square. \quad (4.12)$$

Proof: exterior--exterior

The two estimates in (3.4) give

$$\int_{B_R} G_e \sqrt{U_e} \, dx \leq CR^{-3/2} G_\gamma(t) U_\gamma(t)^{1/2}. \quad (4.13)$$

After integration in s and t ,

$$\mathcal{K}_{D,ee}^\square \leq C\theta(\mathcal{U}_{\text{ext}}^{\infty,\square})^{1/2} \mathcal{D}_{\text{ext}}^\square. \quad (4.14)$$

Adding (4.7), (4.10), (4.12), and (4.14) proves (4.2). $\square$

Remark 4.2 — sharper coupled tails

The separated pair in (2.5) is convenient but not the only bookkeeping. The three exact Hölder pairings produced by the proof are

$$\begin{aligned}
\mathcal{T}_{ce} &= \frac{\nu}{R} \int_I \left(\int_{B_{2R}} |\nabla u|^2 \right) \left(\frac{U_\gamma(t)}{R} \right)^{1/2} dt, \\
\mathcal{T}_{ec} &= \frac{\nu}{R} \int_I G_\gamma(t) \left(\frac{1}{R} \int_{B_{2R}} |u|^2 \right)^{1/2} dt, \\
\mathcal{T}_{ee} &= \frac{\nu}{R} \int_I G_\gamma(t) \left(\frac{U_\gamma(t)}{R} \right)^{1/2} dt.
\end{aligned}
\tag{4.15}$$

Each is scale invariant. They give a sharper mixed-tail formulation if a later argument can estimate them directly. This note does not claim that the separated tails in (2.5) are optimal among every possible coupled functional.

Corollary 4.3 — interface with the pressure-cutoff row

Assume in addition that (u, p) is a periodic suitable weak solution on $\mathbb{T}^3 \times (0, T)$, with $p \in L^{3/2}_{t,x}$, and that the common Ro.73X pressure-tail quantifiers hold. Let

$$Q_s = P_s(pu) - P_s p P_s u, \tag{4.16}$$

let $s : I_R^\square \rightarrow (0, \theta R^2]$ be measurable, and let $\eta_R \in W^{1,\infty}_0(B_R)$ satisfy $\|\nabla \eta_R\|_\infty \leq C_\eta/R$. Combining Theorem 4.1 with the Ro.73X pressure-covariance estimate (5.7) and its local-energy payment (6.2) gives

$$\begin{aligned}
&\mathcal{K}_D^\square(z_0, R; \theta) + \frac{1}{R} \int_{I_R^\square} \int_{B_R} |Q_{s(t)} \cdot \nabla \eta_R| \, dx \, dt \\
&\leq C_{\mathbb{T}^3, \theta, \nu, C_\eta} \left\{ \left[\mathcal{E}^\square(z_0, 4R) + \mathcal{U}_{\text{ext}}^{\infty, \square}(z_0, R; \theta) + \mathcal{D}_{\text{ext}}^\square(z_0, R; \theta) \right]^{3/2} + \mathcal{A}_{\text{ext}}^\square(z_0, R; \theta) \right\}.
\end{aligned}
\tag{4.17}$$

The new velocity endpoint tail and the reused gradient tail pay $\mathcal{K}_D$. The older $\mathcal{A}_{\text{ext}}$ continues to pay the nonlocal pressure and harmonic rows. No claim is made that the two quadratic tails alone control a general Q_s .

05 / CANONICAL REPORT

Why the older exterior functional is insufficient

The Ro.73X functional is

$$\mathcal{A}_{\text{ext}}^\square = \mathcal{G}_{u,p}^\square + \mathcal{H}_u^\square, \tag{5.1}$$

where $\mathcal{G}_{u,p}$ contains the Gaussian annular integral of

$$|u|^3 + |p - c_R(t)|^{3/2}, \tag{5.2}$$

and $\mathcal{H}_u$ contains an algebraically weighted annular $|u|^2$ moment used for harmonic pressure. Neither term contains $|\nabla u|^2$. Also, the Gaussian $|u|^3$ row is integrated in physical time and does not control an essential time supremum of annular L^2 energy.

The next examples are pure function-level obstructions. They are not Navier--Stokes solutions and do not construct a singularity.

Example 5.1 — exterior high-frequency packet

Choose a ball $(B \setminus \text{Subset } A_{2(R)})$ whose closure is disjoint from B_{4R} , and choose $(\phi \in C_c^\infty(B \setminus \text{Subset } A_{2(R)}))$, $\phi \neq 0$. For integers N and amplitudes $\varepsilon_N > 0$, let

$$b_N(y) = \frac{\varepsilon_N}{N} \phi(y) \sin(Ny_1) e_3, \qquad w_N = \nabla \times b_N. \tag{5.3}$$

Regard the packet as a periodic field in the fixed chart and set its associated Poisson pressure to

$$p_N = \mathcal{R}_i \mathcal{R}_j (w_{N,i} w_{N,j}), \qquad -\Delta p_N = \partial_i \partial_j (w_{N,i} w_{N,j}). \tag{5.3a}$$

Then w_N is smooth, divergence free, and supported in B_* . For all sufficiently large N ,

$$\|w_N\|_{L^2} + \|w_N\|_{L^3} \simeq_\phi \varepsilon_N, \qquad \|\nabla w_N\|_{L^2} \simeq_\phi \varepsilon_N N, \qquad \|p_N\|_{L^{3/2}} \leq C_\phi \varepsilon_N^2. \tag{5.4}$$

The last estimate is the periodic Calder--Zygmund bound. For a time-independent packet on a fixed interval,

$$\mathcal{E}^\square(z_0, 4R) = 0, \qquad \mathcal{A}_{\text{ext}}^\square \leq C \varepsilon_N^3. \tag{5.4a}$$

Fix $0 < \alpha < \beta \leq \theta$. The periodic heat kernel has a positive minimum on $s \in [\alpha R^2, \beta R^2]$. Since a periodic curl and its gradient have zero spatial mean, the exact variance formulas give, uniformly for $x \in B_R$ on that scale band,

$$k_s[w_N](x) \geq c \varepsilon_N^2, \qquad D_s[w_N](x) \geq c \varepsilon_N^2 N^2. \tag{5.4b}$$

Consequently

$$\mathcal{K}_D^\square[w_N] \geq c \varepsilon_N^3 N^2. \tag{5.4c}$$

Taking $\varepsilon_N = N^{-2/3}$ yields

$$\mathcal{K}_D^\square[w_N] \geq c, \quad \mathcal{E}^\square(z_0, 4R) = 0, \quad \mathcal{A}_{\text{ext}}^\square = O(N^{-2}) \longrightarrow 0. \tag{5.4d}$$

Thus the older $(\mathcal{E}^\square)^{3/2} + \mathcal{A}_{\text{ext}}^\square$ payment does not control $\mathcal{K}_D$ for arbitrary periodic energy-class velocity/Poisson-pressure pairs. The static packets are not unforced Navier--Stokes trajectories, so (5.4d) is not a suitable-weak NSE counterexample.

Example 5.2 — exterior time concentration

Let $w \in C_c^\infty(B_*; \mathbb{R}^3)$ be nonzero and divergence free. Choose a time interval $J_\delta \subset I$ of length δ , and smoothly approximate

$$w_\delta(t, y) = \delta^{-1/3} 1_{J_\delta}(t) w(y). \tag{5.5}$$

Then

$$\int_I \int |w_\delta|^3 \, dy \, dt = \|w\|_3^3, \quad \operatorname{ess\,sup}_{t \in I} \int |w_\delta|^2 \, dy = \delta^{-2/3} \|w\|_2^2. \tag{5.6}$$

With the associated pressure $p_\delta = \mathcal{R}_i \mathcal{R}_j (w_{\delta,i} w_{\delta,j})$, the complete old $\mathcal{A}_{\text{ext}}$ remains bounded by Calder\'on--Zygmund, while $\mathcal{U}_{\text{ext}}^\infty \rightarrow \infty$. This shows why the old Gaussian velocity tail does not supply the time endpoint used in core--exterior estimate (4.10).

The sequence (5.5) has no uniform $L_t^\infty L_x^2$ bound. It does not contradict energy-class finiteness for any fixed field. If a later theorem allows a global Leray energy on its right-hand side, that global energy can pay this time supremum, but the resulting estimate is no longer a closure by the old local-energy and $\mathcal{A}_{\text{ext}}$ package alone.

Proven rows and remaining gates

PROVED

- The positive four-block majorization (3.2) is dimensionally and algebraically valid.
- The four blocks satisfy (4.7), (4.10), (4.12), and (4.14), with the displayed $\theta^{1/4}$ or θ factors.
- For both matching clocks, the full observable satisfies the local energy/tail bound (4.4).
- The pressure-cutoff row interfaces with the new estimate through the inherited Ro.73X payment, giving (4.17).
- The all-energy-field packets (5.3)--(5.4d) prove that the old $(\mathcal{E}^\square)^{3/2} + \mathcal{A}_{\text{ext}}^\square$ right-hand side alone does not control $\mathcal{K}_D$ in that larger class.
- All quantities in (4.2) are invariant under Navier--Stokes scaling.

FINITE

For every periodic energy-class velocity on the stated interval, $\mathcal{U}_{\text{ext}}^{\infty, \square}$ and $\mathcal{D}_{\text{ext}}^{\square}$ are finite. The first is the one new endpoint tail; the second is the favorable gradient tail already frozen in R0.73X. This is a size statement, not a uniform local bound.

OPEN

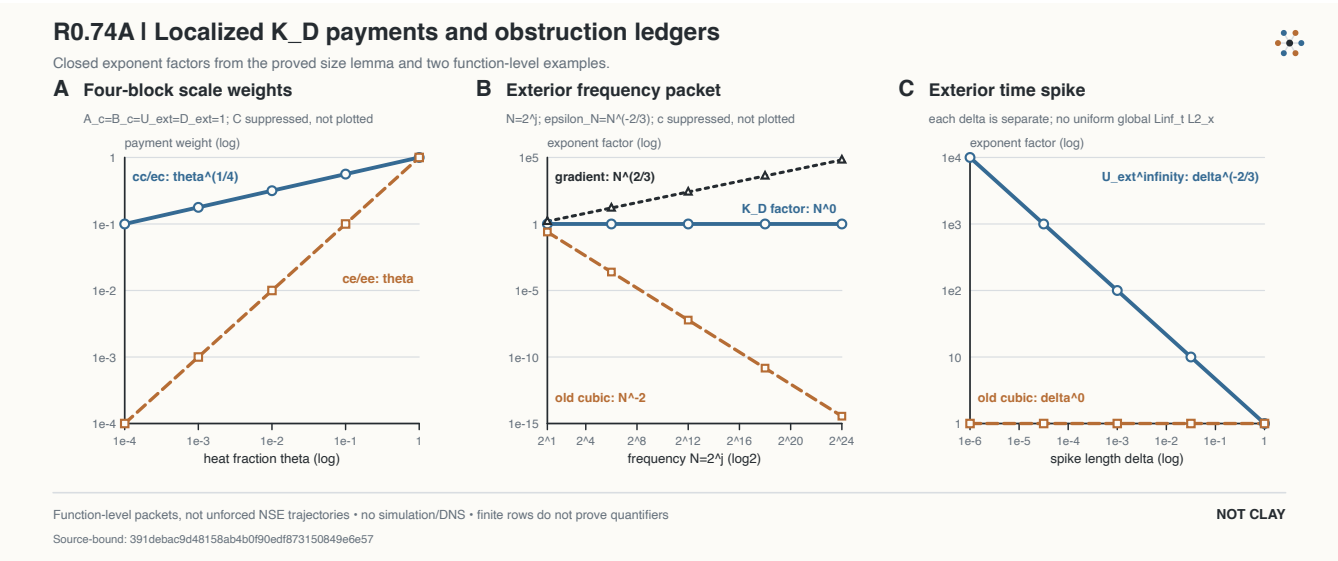
1. Control of the new velocity endpoint tail and the reused gradient tail by data on one smaller cylinder.
2. Smallness or absorption of either quadratic tail in a local energy inequality.
3. A sharper argument replacing the time supremum by a coupled tail that is stable under the intended blow-up sequence.
4. Whether the Navier--Stokes equation supplies a stronger closure absent for arbitrary energy-class velocity/Poisson-pressure pairs; the static packet obstruction does not decide this.
5. Weak stability and lower semicontinuity of the localized observable and its chosen exterior payments.
6. Compatibility of this upper bound with a scale-uniform lower bound after quotienting the precise first-jet near-kernel.
7. Any epsilon-regularity consequence.

NOT CLAY

Nothing in this note proves compactness, epsilon regularity, smoothness, or global regularity for three-dimensional Navier--Stokes solutions.

F / JOURNAL FIGURE

四块付款、旧外部包障碍与 time-supremum tail



PROVED: positiveFourBlockMajorization=PROVED_ANALYTICALLY;
clockMatchedLocalEnergyTailBound=PROVED_ANALYTICALLY;
pressureCutoffInterface=PROVED_BY_INHERITANCE_AND_ANALYTIC_COMBINATION;
oldExteriorPackageOnlyControl=FALSE_BY_EXACT_ENERGY_CLASS_PACKETS;
navierStokesScaling=PROVED_ANALYTICALLY

FINITE: velocityEndpointTail=FINITE_FOR EVERY_STATED_PERIODIC_ENERGY_CLASS_FIELD;
gradientTail=FINITE_AND_IDENTICAL_TO_Ro73X_D_EXT;
localizedKDCertificate=FINITE_ARITHMETIC_CROSS_CHECK_ONLY;
formalEvidenceCertificate=SOURCE_COMMIT_BOUND_PACKAGE_HASH_SEALED;
formalFigurePackage=SEALED_COMMIT_BOUND_25_FILES; formalFigureRows=266;
navierStokesSimulation=NOT_RUN; directNumericalSimulation=NOT_RUN;
ordinaryTranslationPath=LOCAL_DIRECT_NO_DGX; dgxUsed=false

OPEN: smallerCylinderTailControl=OPEN; tailSmallnessOrAbsorption=OPEN; coupledTailReplacement=OPEN;
weakStabilityAndLowerSemicontinuity=OPEN; scaleUniformQuotientCoercivity=OPEN; epsilonRegularity=OPEN;
arbitraryThreeDimensionalGlobalRegularity=OPEN; clayConclusion=OPEN

本节只证明一个正尺度 size lemma: core--core 由 clock-matched local energy 支付, 其余三块由显式 exterior
velocity/gradient tails 支付。它不证明尾项小、可吸收、紧致, 也不证明 epsilon regularity、奇性或任意三维全局正则性。
NOT CLAY。

冻结证明、审计与证书

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